

ACCESS CONTROL LIST

An **Access Control List (ACL)** is a set of rules that is used to filter network traffic and control access to and from a network. ACLs are used on network devices such as routers, switches, and firewalls to permit or deny traffic based on various criteria, such as IP addresses, protocols, port numbers, and more.

There are two types of ACL:

1. Standard ACL (**Range:** 1–99 and 1300–1999)
2. Extended ACL (**Range:** 100–199, 2000–2699)
3. Named ACLs: No numeric range (alphanumeric names)

Named ACL:

Named Access Control Lists (ACLs) provide a more flexible and user-friendly way to manage ACLs compared to the traditional numbered ACLs. With named ACLs, you can assign descriptive names to ACLs, making them easier to understand and manage. Named ACLs also allow for additional functionalities like editing individual entries without deleting the entire ACL, and supporting both standard and extended ACL types within the same configuration mode.

Benefits of Named ACLs:

Descriptive Naming: Named ACLs allow you to use meaningful names instead of numeric identifiers, making ACLs easier to manage and understand.

Edit Capability: Unlike numbered ACLs, named ACLs can be modified by adding or removing specific entries without deleting the entire ACL.

Enhanced Features: Named ACLs provide better control and flexibility, including support for additional match criteria and options.

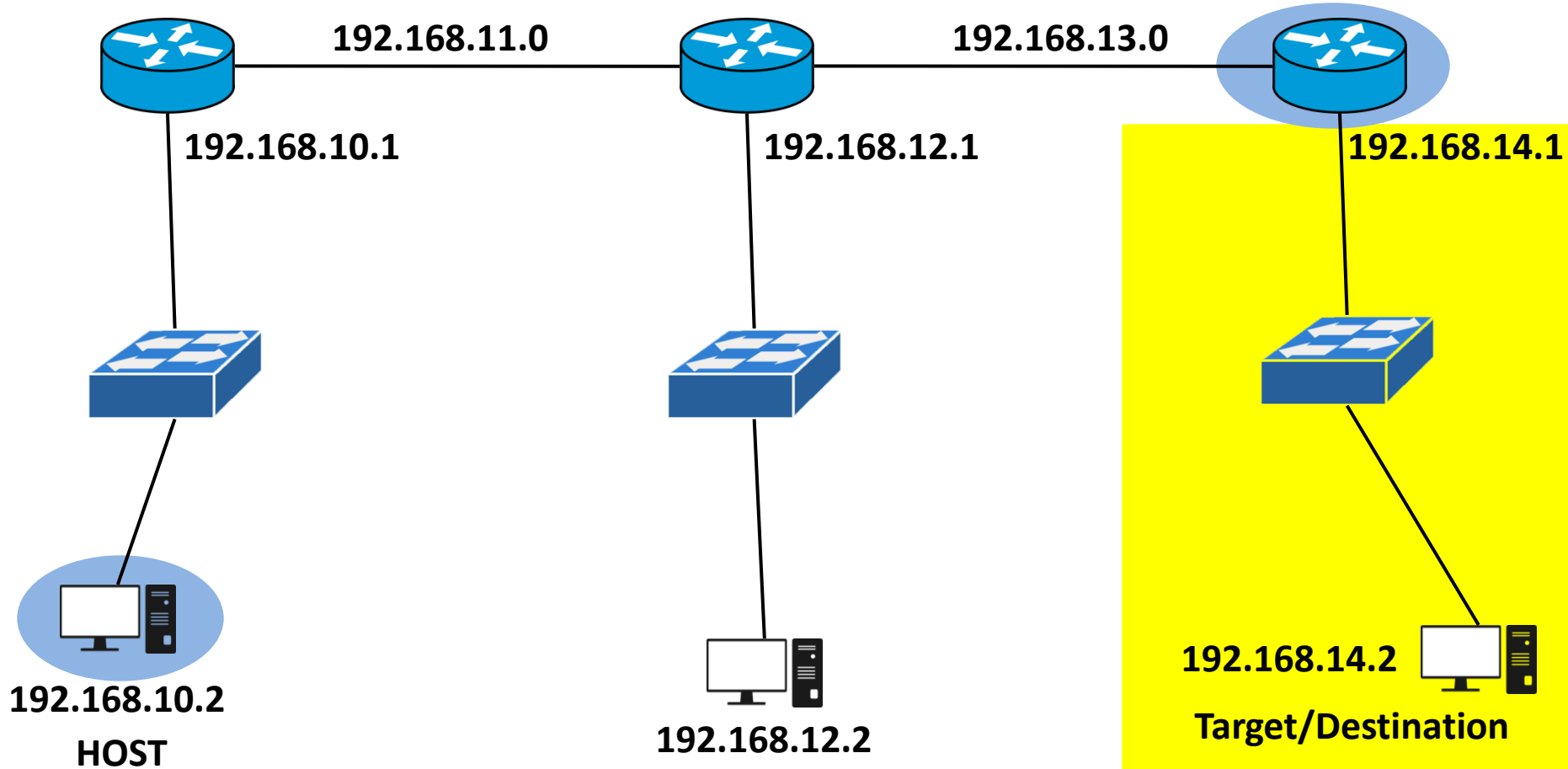
Types of Named ACLs:

Standard Named ACLs: Filters traffic based solely on the source IP address.

Extended Named ACLs: Provides more granular control by filtering traffic based on source and destination IP addresses, protocols, port numbers, etc.

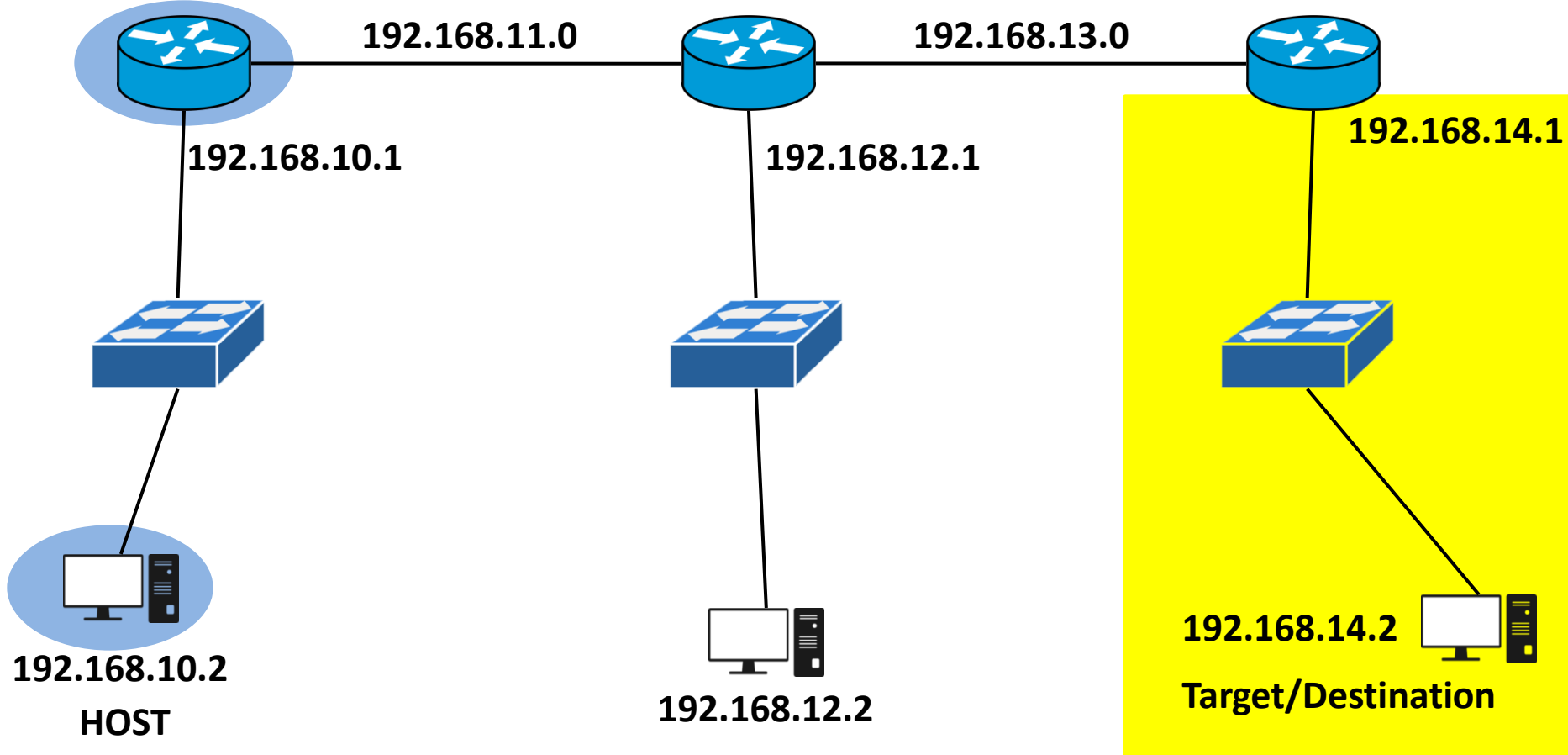
Standard ACL:

Configured on node Closest to the Target/Destination



Extended ACL:

Closest to the HOST/Source



Configuration:

Named Standard ACL:

Router(config)# ip access-list standard ACL_NAME

Router(config-std-nacl)# permit 192.168.10.0 0.0.0.255

Router(config-std-nacl)#deny any

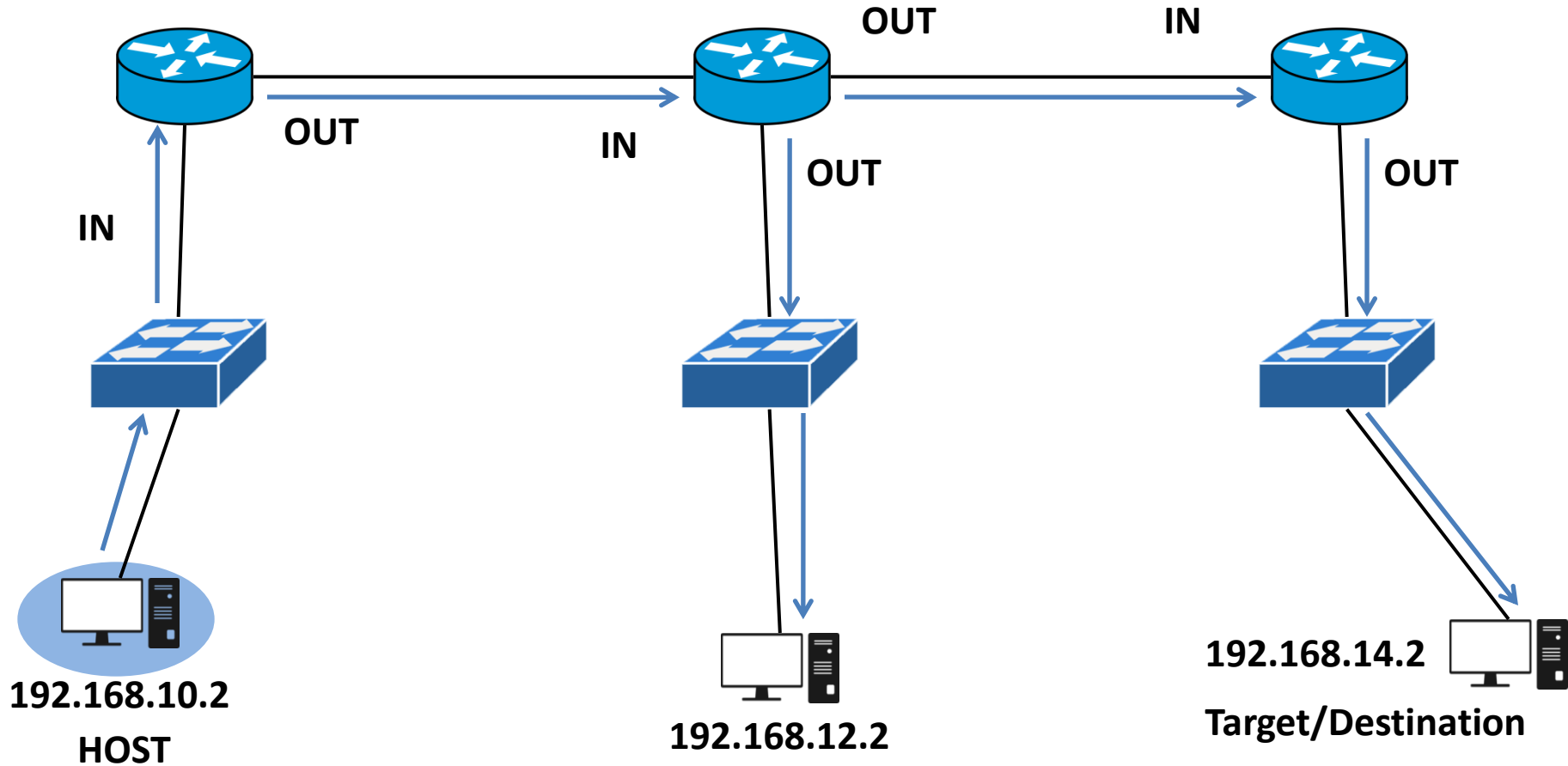
Named Extended ACL:

Router(config)# ip access-list extended WEB_TRAFFIC

*Router(config-std-nacl)# permit tcp 192.168.10.0 0.0.0.255
172.16.20.0 0.0.0.255 eq 80*

Router(config-std-nacl)#deny ip any any

How to decide inbound or Outbound:



Applying ACLs:

Inbound ACLs: Applied to incoming traffic on an interface

Router(config)#interface GigabitEthernet0/0

Router(config)#ip access-group 100 in

Outbound ACLs: Applied to outgoing traffic on an interface.

Router(config)# interface GigabitEthernet0/1

Router(config)# ip access-group 100 out